

Lab-5: CS-2203 Artificial Intelligence Lab

Assignment No: 05

Date: Feb 05, 2026

Submission Deadline: [Feb 05, 2026 - 17:50 Hrs. IST]

Instructions for Submission

General Instructions

Some assignments may need to be submitted via Google Form. Please carefully follow the instructions below to ensure proper submission:

File Preparation

Submission Instructions:

1. Store all assignment-related files in a single folder and compress it into a `.rar` or `.zip` file. The compressed file must be named in the following format:

`roll-number_assign5.rar` or `roll-number_assign5.zip`

2. For example, if your roll number is 2201AI01, the compressed file should be named as:

`2201AI01_assign5.rar` or `2201AI01_assign5.zip`

3. Save each program using the filename format specified alongside each question.
4. If all assignments are completed within a single Jupyter Notebook, name the notebook file using the following format:

`CS2203_Lab5.ipynb`

5. When using a single notebook for multiple problems, clearly **segregate the code for each problem**. Each problem must be introduced using a **text cell** containing the problem statement before the corresponding code segment.
6. After completing the notebook, place it inside the assignment folder, compress the folder, and upload it by following the above naming instructions.

Submission Link

Upload your compressed file using the following Google Form link:

<https://forms.gle/LKHJ1fcDDHUoNrUY9>

Problem Statement 1: Use Breadth-First Search in a One-Way Mini Campus [20 + 10 points]

Scenario

A mini university campus is represented as a 3×3 grid. A student must travel from the **Hostel (S)** to the **Exam Hall (E)**. Due to temporary construction, some walking paths have become **one-way routes**. The student must obey these directional constraints while moving.

Your task is to print the **Traversal Order** and **Shortest Path** determining the **minimum number of steps** required to reach the exam hall using the Breadth-First Search (BFS) algorithm.

If the destination cannot be reached, print **-1**.

Movement Rules

- The student may move Up, Down, Left, or Right.
- Each move counts as one step.
- Direction symbols apply when **leaving the cell**.
- Cells marked '#' are blocked.
- You cannot move out of bound.

Grid Symbols

Symbol	Meaning
S	Start (Hostel)
E	Exam Hall
.	Open path (all directions allowed)
#	Blocked cell
R	Move only Right
L	Move only Left
U	Move only Up
D	Move only Down

Sample Input with Grid Visualization

	0	1	2
0	S	.	R
1	.	#	D
2	.	.	E

Sample Output

Traversal Order: [(-, -), (-, -), (-, -), (-, -), (-, -), (-, -), (-, -)]

Shortest Path: [(-, -), (-,-), (-, -), ?, ?]

Minimum steps: [your answer]

Evaluation Criteria : 1

The evaluation of this lab assignment will be based on the following criteria:

- (a) **Correctness of Implementation (40%):**
- (b) **Code Quality & Q&A (30%):**
 - The code should be well-structured, properly commented, and follow standard Python programming practices.
 - Students may be asked questions related to their implementation to assess their understanding.
- (c) **Error Handling (10%):**
 - The code gracefully handles potential errors, such as file not found or invalid image formats.
- (d) **Documentation (10%):**
 - The code is accompanied by appropriate documentation and comments that explain the steps taken in each part of the program.
- (e) **Timeliness of Submission (10%):**
 - The assignment is submitted on time, adhering to the submission deadline.

Evaluation Criteria: 2

The evaluation of this lab assignment will be based on the following criterion:

- (a) **Viva (10 Marks):**
 - Questions will be asked related to the assignment to assess the student's understanding of the implemented concepts.